

Pug & Pugsan — Step-by-Step Build Guide

For Sal / PapaThaiGG · Personal stream co-host first · Updated August 10, 2026
Characters: **Pug** (Android / ALPHA VRM) → then **Pugsan** (samurai pug humanoid)
Home chat: Cursor project PapaThaiGG (Mozart + Sal)
Open first for status: Desktop 00 - PUG PROJECT MASTER GUIDE (Sa1).pdf
Related: PUG_AI_COHOST_CHECKLIST.md (full checklist) · this file = ordered build steps

What this guide is: Exact build order — brain → body on stream → chat controls → dual-PC → Pugsan → sell later.
What it is not: The live status sheet (that’s the Master Guide PDF). Sell-later product ideas live in AI_Stream_Cohost_App_Syllabus .

0. Dual-PC hardware map (locked in)

	Gaming PC	Streaming PC (Pug home)
Job	Play the game clean — max FPS, capture out to Elgato	Pug / Pugsan / AI / Warudo / OBS / Steam (Warudo) — all co-host work lives here
CPU	Intel Core i9-12900KF	Intel Core i9-10900KF
RAM	32 GB DDR5 (upgrade later → 64 GB DDR5 when budget allows)	64 GB DDR4
GPU	NVIDIA GeForce RTX 3080 Ti	NVIDIA GeForce RTX 3080
Hostname	DESKTOP-JCVJ6HR	RANDYS_PC_TH
Capture	HDMI/DP out → Elgato	Elgato 4K60 in · OBS composites game + Pug

Important: Early brain work started on the *gaming* PC (D:\PTGG\). Pug’s live home is now the **streaming PC** (OneDrive\Desktop\App ideas\PTGG\). Gaming PC should not fight the game for VRAM once dual-PC is live.

1. Is Ollama the right brain? (Yes — for you)

Choice	Cost	Verdict for PapaThaiGG
Ollama + local model (what we use)	Free (3080 / 3080 Ti you already own)	Best path. No monthly bill. Private. Fast enough on your GPU. Personality = Modelfile text you control. Final home = streaming PC .
Ollama + Dolphin-style models	Free	Optional later. Often less filtered / more “chaotic funny.” Riskier on live stream. Try after Pug works with llama3. Keep a stream-safe Pug backup.
Cloud APIs (OpenAI, etc.)	Pay per use	Not needed now. Useful later as an <i>optional</i> tier for customers without a strong GPU.

You do **not** need to spend hundreds or thousands on AI APIs to build this. Hardware you already have + free Ollama is the correct MVP path. When selling later: local brain by default, optional cheap cloud for people without GPUs — and a simple “Attitude” slider that rewrites Modelfile personality.

2. Where you are right now (Aug 10, 2026)

- ✓ Blender 5.2 · ALPHA.vrm · Warudo scene **Pug (Android)** saved on streaming PC
- ✓ Ollama · llama3 · custom Pug · wholesome Modelfile · identity fix (speaks as Pug)
- ✓ Python 3.12 + Pug_Brain.py · Desktop Talk to Pug.bat · Piper TTS auditioning
- ✓ VB-Cable installed · OBS + Warudo · Pug voice via CABLE + Monitoring
- ✓ Pugsan T-pose / reference art on disk (build later — Phase H)
- → Next (Phase C): lock Piper voice · mic talk · chat → Pug · mute · walk on/off · scene presets

3. Step-by-step — do in this order

1 Talk to Pug (brain only) — DONE

Easiest: Desktop → double-click Talk to Pug.bat

Or PowerShell from PugAI:

```
cd "$env:USERPROFILE\OneDrive\Desktop\App ideas\PTGG\PugAI"  
& "$env:LOCALAPPDATA\Programs\Python\Python312\python.exe" .\Pug_Brain.py
```

- Type to him · he answers + TTS speaks
- Type quit to stop

2 Tune personality (Modelfile) — DONE (wholesome locked)

- ✓ Wholesome / warm / family-friendly · 1–2 short sentences · Papa nicknames (never "Sal")
- Rebuild anytime: from PugAI folder → ollama create Pug -f Modelfile
- *Optional later:* Dolphin base for edgier comedy · keep stream-safe Pug as backup
- *Still tuning:* nickname usage percentages · Piper voice lock

3 Install VB-Cable (audio pipe for stream) — DONE

- ✓ CABLE Input / CABLE Output in Windows Sound
- ✓ Python → CABLE Input · OBS → CABLE Output · Monitoring Enabled · USB headset default
- **Never** set Windows default output to CABLE Input

4 Warudo body — Phase B — DONE

- ✓ Scene **Pug (Android)** · transparent BG · Orbit camera · Focus Character
- Optional polish: Idle animation · MediaPipe Tracker stays idle (Face/Hand off · Pug Motion Capture = No)
- Keep Pug small in frame — corner mascot, not full-screen

5 OBS — put Pug on the canvas — MOSTLY DONE

- ✓ Window Capture → Warudo · Pug Voice = Audio Input Capture → CABLE Output
- Scene presets still to build: **Cam+Game+Pug · Game+Pug · Game only · Cam+Game**
- Optional: crop/pad Warudo edges · Spout2 later

6 Mouth / expressions follow speech — LATER POLISH

- Wire TTS audio (or Warudo lip-sync) so mouth moves when Pug talks
- Map expressions to mood later (happy / hype / gentle)
- Goal: viewers see + hear the co-host, not a frozen statue

7 Stream controls (co-host manners) — PHASE C

- Hard mute: “Pug, be quiet until I call you back”
- Walk on → talk → walk off (not always on screen)
- Hotkeys first; Stream Deck as backup

8 Chat → Pug (Twitch / YouTube) — PHASE C

- Streamer.bot on Desktop → `Streaming\Streamer.bot-x64-1.0.4\` — good bridge candidate
- Chat message → filter → Ollama Pug → TTS → avatar
- Rate limits so he doesn’t spam

9 Dual-PC hookup — PHASE D

- **Gaming PC:** game only → video out to Elgato
- **Streaming PC:** Elgato in + OBS + Warudo + Ollama Pug + TTS + VB-Cable
- Cable: game GPU HDMI/DP → Elgato 4K60 → streaming PC
- Later upgrade: gaming PC RAM 32 → 64 GB DDR5 (nice, not required for Pug)

10 Pugsan — PHASE H (after Pug is fun on stream)

- T-pose art → 3D (Tripo or similar) → Blender rig / mouth shapes → `Pugsan.vrm`
- Load in Warudo · same Ollama brain · swap bodies for variety
- **Do not** block Pug’s first live streams waiting on this

11 Only after proof — think about selling — PHASE I

- Package control app + “Attitude” personality editor (Modelfile made friendly)
- BYO avatar import for customers · Official Pug / Pugsan packs as your IP
- Pricing later: one-time vs monthly — after beta streamers tell you what they’d pay

4. Important paths

What	Where (streaming PC — current home)
Talk to Pug	Desktop → <code>Talk to Pug.bat</code>
Pug brain	<code>OneDrive\Desktop\App ideas\PTGG\PugAI\</code> (Modelfile, Pug_Brain.py, voices)
ALPHA VRM	<code>...\PTGG\ALPHA.vrm</code>
All PTGG files	<code>...\Desktop\App ideas\PTGG\</code>

Streamer.bot	Desktop → Streaming\Streamer.bot-x64-1.0.4\
Master guide PDF	Desktop → 00 - PUG PROJECT MASTER GUIDE (Sal).pdf
This step-by-step	Repo docs\PUG_PUGSAN_STEP_BY_STEP_BUILD · Plans folder copy · Desktop PDF
Checklist (repo)	PapaThaiGG\docs\PUG_AI_COHOST_CHECKLIST.md

Gaming PC archive paths (early build — don't use as live home): D:\PTGG\PugAI\ · D:\PTGG\ALPHA.vrm · D:\PTGG\Blender Pug and Pugsan avatar builds\Pug 1.blend · D:\PTGG\PapaThaiGG brand build\...\PapaThai Mascot Files\ · D:\PTGG\VBCABLE_Driver_Pack45\

5. Your next session (Phase C prep)

- 1. Lock one Piper voice for stream (Northern English leading · finish Joe/Norman audition)
- 2. OBS scene presets: Cam+Game+Pug · Game+Pug · Game only
- 3. Then: mic talk-to-Pug · chat → Pug · mute · walk on/off

FFmpeg / Node = optional · Elgato = Phase D · Fun first, retired pace · Mozart + Sal
Synced into GitHub docs/ · PapaThaiGG Plans · Desktop — safe to delete the old "Cursor work info" folder.